SmartGlove System Overview

Non-Technical Summary, Thresholds, and Decision Making

This document provides a clear, high-level summary of how the entire SmartGlove ecosystem operates. It breaks down the system into core components and explains exactly how the software makes automated health decisions.

1. Hardware & Physical Sensors

The system relies on physical wearable devices that sit directly on the user's body to collect continuous health data.

The Smart Glove (Wrist/Hand)

The glove is the primary device. It constantly reads pulse, blood oxygen (SpO2), skin temperature, hand movement, and sweat levels. It also has a built-in GPS.

Decision / Thresholds:

- **Finger Detection:** The glove uses an infrared light threshold to detect if it is actively being worn. If no finger is detected, it enters a 'Standby' mode to save battery.
- **SOS Shake Gesture:** The glove looks for sudden, violent movements. If it detects **3 firm shakes (> 18.0 m/s²) within a 2-second window**, it overrides all other rules, turns the OLED screen into an emergency display for 10 seconds, and immediately blasts an SOS signal.

The Smart Belt (Waist/Spine)

A secondary companion device worn on the waist. It strictly tracks posture and torso impact.

Decision / Thresholds:

- **Fall Detection:** If the belt experiences a sudden physical impact followed by a sustained lack of movement, it registers a 'Fall'.
- **Posture Tracking:** By measuring tilt angles, it categorizes the wearer into Standing, Sitting, Walking, Running, or Lying Down.

2. The Local Hub (Brain of the System)

Instead of sending heavy, raw sensor data straight to the internet (which drains battery and requires a strong signal), the glove sends data locally to a nearby computer, phone, or Raspberry Pi acting as the 'Hub'.

What it does:

- **Cross-Checking (Sensor Fusion):** It combines data from both the Glove and the Belt to make smarter decisions. For example, a fast heart rate while 'Running' is ignored, but a fast heart rate while 'Lying

Down' is flagged as a risk.

- **Live Dashboard:** It hosts a local web monitor where users or nearby caretakers can watch vitals in real-time.

3. Health Screening & Decision Logic

The hub uses software 'engines' to constantly screen for specific health risks. Each engine outputs a likelihood percentage from 0% to 100%.

A. Hypoxemia (Low Blood Oxygen)

Monitors the SpO2 saturation to ensure the brain and body are getting enough oxygen.

- **Normal (Safe):** 95% to 100%
- **Warning:** Below 95% (Triggers elevated risk)
- **Critical:** Below 90% (Triggers immediate emergency)
- **Silent Hypoxia Engine:** Even if oxygen is 'Normal' right now, the system calculates the 'trend' over the last few minutes. If it predicts oxygen will crash below 85% soon, it alerts the user early.

B. Arrhythmia & Heart Irregularities

Monitors the heart rate (BPM) for abnormal speeds or skipped beats.

- **Bradycardia (Too slow):** Drops below 60 BPM for several consecutive readings.
- **Tachycardia (Too fast):** Rises above 100 BPM while the user is resting.
- If the heart rate spikes and drops chaotically within a few seconds, the engine flags a high likelihood of irregular beats (Arrhythmia).

C. Parkinson's / Resting Tremor

Analyzes the micromovements of the hand.

- **Decision Logic:** The engine filters out normal arm swinging. It specifically looks for a rhythmic shaking at exactly **4.5 to 5.0 vibrations per second (Hz)**. If this specific frequency is detected while the hand is otherwise still, the Parkinson's likelihood score goes up.

D. Sepsis (Systemic Infection)

Sepsis is deadly because it attacks the whole body. The system uses a medical rule called SIRS (Systemic Inflammatory Response Syndrome) to detect it.

- **Decision Logic:** It triggers if multiple vitals fail at the exact same time: a fever (Skin Temp > 38°C) combined with a racing heart (HR > 90 BPM).

E. Acute Stress & Anxiety

Looks at the nervous system's 'fight or flight' response.

- **Decision Logic:** If a user's sweat levels spike (measured by Galvanic Skin Response/GSR) and their heart rate elevates, but the belt confirms they are just sitting still, the system identifies this as a psychological stress event rather than exercise.

4. The Master Risk Score & Alerts

To avoid spamming caretakers with confusing medical data, the system boils all the AI engines down into one simple number: The Master Risk Score (0 to 100).

Risk Score	Status	System Action
0 to 30	NORMAL	Does nothing. Vitals are healthy.
31 to 60	ELEVATED	Highlights the dashboard in yellow. Keeps a closer eye on trends.
61 to 100	CRITICAL	Flashes red. Triggers global Emergency Alert.

How the Final Alert Decision is Made:

An emergency notification (including the user's live GPS coordinates) is immediately sent to the cloud (Firebase) and dispatched to mobile phones if ANY of the following 3 conditions are met:

1. The Master Risk Score exceeds 60.
2. The Smart Belt detects a hard Fall.
3. The user performs the physical SOS Shake Gesture with the glove.

If the emergency is resolved (e.g., the vitals return to normal, or a nurse clicks 'Dismiss Alert'), the system resets and returns to quiet monitoring.